

Nevil J

Coimbatore, India

nevilj22@gmail.com — +91-9342886761

 [linkedin.com/in/nevilj](https://www.linkedin.com/in/nevilj) —  github.com/nevil2006 —  nevil-jportfolio.vercel.app

Summary

AI/ML Engineer specializing in Computer Vision, Vision-Language Models, and real-time AI systems. Experienced in building low-latency pipelines using YOLO, CLIP, and OpenCV for smart city and healthcare applications. Strong focus on edge AI, tracking systems, and scalable deployment.

Education

KGiSL Institute of Technology

2023 – 2027

B.Tech in Artificial Intelligence and Data Science

CGPA: 7.2/10

Skills

- **Languages:** Python, SQL, JavaScript, Embedded C
- **ML/DL:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- **Computer Vision:** OpenCV, YOLO, CLIP, DeepSORT, ByteTrack
- **Frameworks:** Flask, FastAPI, Streamlit
- **Tools:** Git, Docker, Jupyter, Google Colab, Roboflow
- **Systems:** Real-Time Systems, Edge AI, IoT (ESP32, MPU6050)

Projects

AI Traffic Management System (Real-Time CV + IoT)

2024 – 2025

github.com/nevil2006/AI-Based-Smart-Traffic-Management-System

- Built real-time detection + tracking pipeline achieving **92% accuracy at 30 FPS**
- Designed adaptive signal control reducing congestion by **35%**
- Implemented DeepSORT tracking with unique ID assignment and violation detection
- Integrated ESP32 edge system for low-latency signal automation

PathoVision: Breast Cancer Classification (CLIP + VLM)

2025 – 2026

github.com/nevil2006/medical-analysis-app

- Developed multimodal classification pipeline using CLIP (Vision-Language Model)
- Achieved **88% F1-score** using prompt engineering and feature alignment
- Applied explainable AI techniques for medical interpretability

Embedded IoT Fall Detection System (Healthcare Safety)

2026

github.com/nevil2006/fall-detection

- Built real-time fall detection system using ESP32 and MPU6050 (sensor fusion)
- Designed motion-analysis pipeline using thresholding for fall detection
- Implemented event-triggered alert system for emergency response
- Optimized system for edge deployment with low-latency performance

ID Document Masking Tool (OCR + CV)

Jun 2025 – Jul 2025

github.com/nevil2006/masking-tool-in-image

- Built OCR-based redaction pipeline for sensitive information detection
- Integrated Tesseract with bounding-box localization
- Improved accuracy using preprocessing (denoising, contrast enhancement)

AI Biometric Authentication System

2025

github.com/nevil2006/biometric-authentication

- Developed face recognition system with liveness detection + OTP verification
- Built secure backend APIs using Flask

Experience

AI Project Intern — Centre for Machine Learning and Intelligence (CMLI)

Jun 2025 – Jul 2025

- Built CLIP-based classification system for medical imaging
- Implemented zero-shot and few-shot learning pipelines
- Designed structured multimodal datasets

International Collaboration

BIOE 396/496 Study Abroad Program — Lehigh University

Mar 2026

- Participated in interdisciplinary research on AgriTech and global innovation
- Collaborated with international teams on lab-based problem solving
- Gained exposure to cross-cultural research workflows and experimentation

Achievements

- 4× Hackathon Winner (National Level)
- Best Team — IFET Hack Warriors 2026
- 1st Place — Trixathon 2025, SNS IdeaThon
- AI Traffic System approved for pilot deployment by Coimbatore Police